Amirmahdi Karambakhsh

Molecular & Cellular Biology | Disease Mechanisms | Integrative Data Analysis

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RESEARCH INTERESTS

Molecular mechanisms of cancer and complex diseases, experimental cell biology, signaling pathway regulation, functional genomics, integrative data-driven approaches combining wet-lab and computational analysis

EDUCATION

Sep 2025 – Jul 2026 (exp.) **Master's degree in Systems and Synthetic Biology**
Université Évry Paris-Saclay, Évry-Courcouronnes, France

Sep 2021 – Aug 2025 **Bachelor's degree in Cell and Molecular Biology**
GPA: 18.06/20 University of Tehran, Tehran, Iran

RESEARCH EXPERIENCE

UMR168 – Physics of Cells and Cancer (PCC), Institut Curie Paris, France
Master's Research Intern *Jan 2026 – Jul 2026 (exp.)*
Supervisor: Dr. Céline Cordier (Team Hersen)

Project: Spatiotemporal Control of Cell Migration

- Investigating how spatiotemporal optogenetic activation of Rac1 modulates cytoskeletal dynamics and collective migration behavior in epithelial cell monolayers.
- Designing and performing live-cell imaging experiments with spatially patterned light stimulation using a custom DMD-based optogenetic system.
- Calibrating patterned illumination and optimizing experimental parameters to ensure reproducible and spatially confined activation.
- Developing quantitative image analysis workflows in Fiji and Python for segmentation, single-cell tracking, and extraction of dynamic migration parameters (velocity, persistence, directionality ratio, mean square displacement).
- Analyzing collective behavior through population-level metrics and correlation analysis to characterize signaling-dependent changes in coordination and migration patterns.
- Integrating experimental observations with quantitative analysis to characterize how localized Rac1 activation influences collective migration dynamics.

School of Biology, University of Tehran Tehran, Iran
Undergraduate Researcher *Aug 2024 – Jul 2025*
Supervisor: Professor Mehriar Amininasab, Associate Professor

Project: Integrative Epigenomic and Network-Based Analysis of Triple-Negative Breast Cancer

- Designed and implemented a bioinformatics workflow to analyze genome-wide DNA methylation profiles in TNBC, processing >750,000 CpG sites.
- Identified differentially methylated regions (DMRs) and performed functional enrichment analysis (GO, KEGG) to characterize pathway-level epigenetic alterations.
- Constructed and analyzed protein–protein interaction networks using STRING and Cytoscape, applying MCODE clustering and cytoHubba centrality measures to identify key regulatory modules and hub genes.
- Integrated epigenomic and network-level analyses to uncover systems-level signatures with potential biomarker relevance in TNBC progression.
- Initial exploratory machine learning approaches were evaluated to assess classification potential prior to focusing on network-based systems analysis.

TECHNICAL SKILLS

Molecular & Cellular Biology:
Mammalian cell culture (human cancer cell lines), sterile techniques, cell counting and viability assays, MTT assay, colony formation assay, apoptosis/necrosis assessment, flow cytometry, DNA/RNA extraction, qRT-PCR, SDS-PAGE, Western blotting, protein purification (kit-based methods).

Live-Cell Imaging & Microscopy:
Time-lapse live-cell imaging and experimental acquisition using MetaMorph; advanced quantitative image analysis using Fiji/ImageJ and custom Python workflows for segmentation, tracking, and dynamic parameter extraction.

Animal Models:
Laboratory mouse and rat handling, basic surgical procedures, disease modeling, anesthesia and post-procedural monitoring, adherence to animal welfare and ethical research standards.

Bioinformatics & Systems Analysis:
DNA methylation analysis, DMR identification, functional enrichment analysis (GO, KEGG), protein–protein interaction network construction and module detection (STRING, Cytoscape, MCODE, cytoHubba).

Computational & Data Analysis:
Python, R, statistical analysis, directed evolution dataset analysis (experimental data), exploratory machine learning approaches, reproducible data workflows; basic familiarity with MATLAB.

Structural Visualization Tools:
UCSF ChimeraX, PyMOL

TEACHING EXPERIENCE

School of Biology, University of Tehran — Tehran - Iran
Teaching Assistant — *Sep 2022 – Aug 2025*

- Assisted in undergraduate courses including Structural Biochemistry, Cell and Molecular Biology I & II, Basic Genetics, Biophysical Chemistry, and Practical Biostatistics.
- Led discussion sessions and problem-solving classes, clarifying complex molecular and quantitative concepts.
- Mentored students individually and supported experimental data interpretation and exam preparation.
- Coordinated and supervised a team of teaching assistants during final academic term.

LANGUAGES

- **English:** Fluent – C1(IELTS 8.0)
- **Persian:** Native
- **French:** Elementary – A1(Currently Improving)

HONORS & AWARDS

- Ranked 2nd in the Cell and Molecular Biology program, University of Tehran (GPA: 18.06/20).
- National Biology Olympiad (Iran) – 15th place nationwide.
- Full undergraduate scholarship, University of Tehran.